The State-of-the-Art Phase-Change Integrated Photonic Devices With Electrically Driven Microheaters

Jing Wen, Jiyuan Jiang, Jinyong Ma, Shuqing Feng, Qingkun Gong, Wanang Xiao, Xiaojuan Lian, and Lei Wang

***Abstract*— Phase-change integrated photonics devices, evolving from the combinations of phase-change materials (PCMs) and silicon photonics, are subjecting to intensive research today. This is because phase-change integrated photonics devices inherit performances superiorities of both phase-change storage memory and silicon photonics in data transmission system. The programming mechanism of phase-change integrated photonic devices can be either optically driven switching or electrically driven switching. In comparison with its optical rival, electrically driven switching, usually realized by an integrated microheater, requires simpler programming scenario and less complex circuits and thus leads to a variety of microheater designs. To more comprehensively understand the physics behind different designs and to distinguish their performances variations, we first reviewed the operational principles of phase-change integrated photonics devices in association with common phase-change photonic materials. The current status of electrically driven switching-based devices is subsequently elaborated according to different microheaters, and their respective merits and drawbacks are interpreted. The technical challenges and future prospects of phase-change integrated photonics devices are eventually discussed.**

***Index Terms*— Electrical, integrated, microheater, phase-change, photonics.**

I. Introduction

PHOTON and electron are fundamental particles in planet, contributing to the constitution of light and electricity, respectively. It is well known that current data transmission system, such as optical fiber, strongly relies on light propagation, while current data storage systems exemplified by static random access memory (SRAM) are governed by electricity conduction. It is therefore natural to conceive an advanced system merging the advantages of electronic storage and optical transmission. Such dream is partially coming true by implementing optical connections among electronic chips and even in interchip data transfer [1]. However, this mandatorily requires the cumbersome electro-optic conversions at the cost of extra energy and time consumption. To address the above issue, all-optical hardware based on silicon (Si) photonics is obviously a promising solution by manipulating data at the speed of light with the exemption of electro-optic conversions [2]. Moreover, Si photonics hardware provides very similar effects to the nonlinear dynamics of neural networks [3], thus making it particularly prospective for neuromorphic applications. In spite of its attractiveness, Si photonics neuromorphic hardware is still facing some stringent challenges, such as the difficulties in achieving storage and logic and the immaturity of photonic manufacturing industries.

In addition to device manufacturing technologies, finding appropriate functional materials plays an important role in determining the performances of future photonic devices. Phase-change materials (PCMs), also known as chalcogenide alloy, usually refer to the composites that include the elements of group 16 of the periodic table and its neighboring groups. The most charming trait of PCMs arises from its ability to exist at two distinct physical states, i.e., amorphous and crystalline configurations. Amorphous structure, corresponding to short-ranged atomic order, usually exhibits large electrical

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Jing Wen is with School of Information Engineering, Nanchang Hangkong University, Nanchang 330063, China, and also with the Jiangxi Hongdu Aviation Industry Group, Nanchang 330000, China (e-mail: wenj@nchu.edu.cn).

Jiyuan Jiang, Xiaojuan Lian, and Lei Wang are with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210023, China (e-mail: 1222228504@njupt.edu.cn; xjlian@njupt.edu.cn; leiwang1980@njupt.edu.cn).

Jinyong Ma is with the School of Information Engineering, Nanchang Hangkong University, Nanchang 330063, China (e-mail: 1600178283@qq.com).

Shuqing Feng is with the Shanghai Radio Equipment Research Institute, Shanghai 201108, China (e-mail: 1216108617@qq.com).

Qingkun Gong is with the Jiangxi Hongdu Aviation Industry Group, Nanchang 330000, China (e-mail: 27992875@qq.com).

Wanang Xiao is with the Institute of Semiconductors, Chinese Academy of Sciences Institute, Beijing 100083, China, also with the Center of Materials Science and Optoelectronics Engineering, School of Integrated Circuits, University of Chinese Academy of Sciences, Beijing 100049, China, and also with Jiangsu JITRI Intelligent Integrated Circuit Design Technology Company Ltd., Wuxi 214028, China (e-mail: waxiao@semi.ac.cn).

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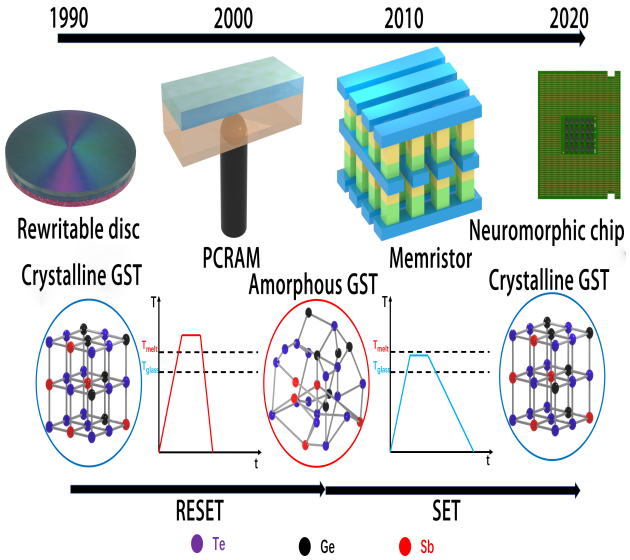


Fig. 1. Temporal evolution of phase-change devices (top) and work modes of the GST media adopted for conventional photonic and electronic applications (bottom).

resistivity and low optical reflectivity, whereas crystalline structure with long-ranged atomic order conversely presents small electrical resistivity and high optical reflectivity. More importantly, these two states can be rapidly and reversibly switched between each other. Heating amorphous state above glass-transition temperature, which is subsequently subjected to slow cooling, can transform PCMs into their crystalline structures, referring to "SET." On the other hand, crystalline phase where temperature is brought up above melting point, followed by a quick quenching, can be transformed to the amorphous phase, also known as "RESET." As physical properties of amorphous state vastly differ from those of crystalline state, it is therefore viable to denote the physical properties of amorphous and crystalline states as binary data "1" and "0," respectively. As a result, both storage and logic computing functionalities can be realized on PCMs-based devices [4].

Owing to its distinguishable physical properties between amorphous and crystalline phases, PCMs, particularly represented by the $Ge_2Sb_2Te_5$ (GST) alloy, triggered the debut and prosperity of phase-change memories at the beginning of the 21st century, represented by phase-change rewritable optical disk [5] and phase-change RAM (PCRAM) [6]. Besides, numerous intermediate states of PCMs can be excitingly obtained by sophisticatedly tailoring the volume ratio of crystalline to amorphous phases [7], implying its multilevel functionality even if encountering power-off. Such nonvolatile trait consequently exploited new applications of PCMs on memristor and neuromorphic computing fields, thereby merging the functionalities of both storage and computing. Fig. 1 graphically presents the historic progress of PCM-based technology mainly in the electronic regime. Inspired by its electronic counterpart, PCMs has most recently been integrated with Si photonics to further boost Si photonics-based devices to an unprecedent height [8]. Note that the stimulus sources for governing the phase transformation of such novel device can be either optical or electrical pulses. A variety of literatures have been published in the past to review the state-

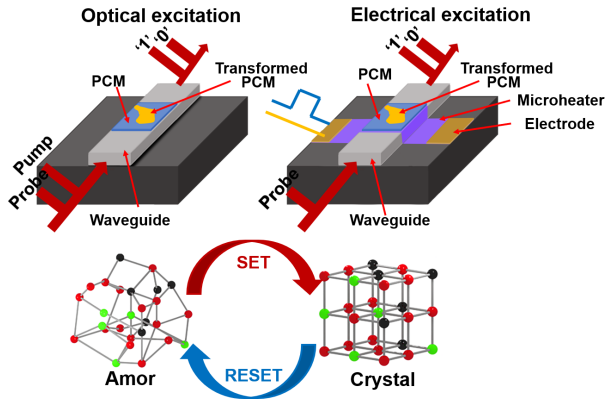


Fig. 2. Schematic illustration of the difference between phase-changed photonic devices induced by optical and electrical excitation (top) and their fundamental working mechanisms (bottom).

of-the-art phase-change integrated photonic devices excited by optical pulses, while those with electrical pulse excitations were rarely reported [9], [10], [11], [12]. All-optical phase-change memory that in fact requires extra electronic components to drive the excitation sources and is usually accompanied with complex optical couplings, nonnegligible interferences, and increased power consumption. In contrast, electrically driven strategies are compatible with CMOS technology, offer better scalability, and simplify packaging. Under this circumstance, here we first reviewed the difference on the operational mechanisms between phase-change integrated photonic devices excited with optical and electrical pulses as well as some common photonic PCMs. The current status of electrically driven switching-based devices is subsequently elaborated according to different microheaters, and their respective merits and drawbacks are interpreted. The technical challenges and future prospects of phase-change integrated photonics devices are eventually discussed.

II. Operational Mechanisms of Phase-Change Integrated Photonic Devices and Photonic PCMs

The operational mechanisms of phase-change integrated photonic devices are schematically interpreted in Fig. 2. Its benchmark structure, originally designed for all-optical photonic memory [13], comprises a PCM layer located on Si or silicon nitride (Si_3N_4) waveguide, which are deposited on a silicon-on-insulator (SOI) platform. To alter the state of PCM, it is necessary to heat the PCM using either optical pulses or electrical pulses to induce the phase transition. Optical pulses are usually transmitted through the photonic waveguide storage unit. When optical pulses are employed, the PCM absorbs light energy and thus generates heat to initiate phase change. In contrast, electrical pulse method induces electric current injected through the external microheaters to trigger phase change via Joule heating. It was reported that a multilevel optically driven waveguide storage device [13] achieved a switching energy of 13.4 pJ and speeds close to 1 GHz. Recent research also demonstrated a 6-bit optically driven waveguide storage unit with over a million cycles of endurance [14]. In spite of its advantages, inherent PCM losses that cause signal attenuation still seem formidable for optical switching.

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In contrast, electrical switching allowed for substantial switching contrasts (300%) with energy consumption in the order of a few nanojoules [15]. Furthermore, electrical switching holds the potential for enabling large-area PCM phase transitions up to 0.4×0.4 mm^2 [16]. Most importantly, electrical switching using microheater is able to program low-loss or lossless PCM, which is a thorny challenge that optical switching struggles to overcome.

All-optical design undoubtedly enables data processing and transport at the speed of light, high modulation speeds, and parallelization due to its unique multiplexing. However, when it is expanded to the system level, extra optical components are required in order to generate and route light, thus imposing a formidable limit on device scalability. Additionally, there are potentially high losses from these optical components, which increases the energy consumption required for programming. An exciting fact is that implementing external local electric heating can provide scalable phase transition with extremely fast switching speeds and fairly low energy consumption [17], previously demonstrated by mature PCRAM technologies. For the above reasons, electrical excitation was most recently regarded as an alternative programming strategy for phase-change integrated photonic device. The device architecture for electrical excitation resembles optical excitation counterpart except that an additional heater layer (also called microheater) is sandwiched between the PCM and waveguide. To induce phase transformation, a programming current applied between two metal electrodes, flows into the microheater, and generates Joule heating at PCM–microheater interface where phase transition occurs. As electric current generation and connection can be realized according to mature and simple techniques, phase-change photonic device with integrated microheater exhibited higher scalability than its photonic rival, and its performances can be mitigated by further optimizing the electrical interface.

Appropriate PCMs for photonic applications with integrated microheaters need to fulfill several requirements. From electrical programming point of view, PCMs require a rapid phase transition targeted for fast write speed. Besides, a relatively low phase-transition temperature is usually preferred to allow for small energy consumption. Moreover, in order to pursue long data retention, PCMs are usually desired to be stable at room temperature. In terms of optical readout, one key requirement is the large optical properties difference between the amorphous and crystalline phases, which can benefit resulting reading contrast and thus enable discernible readout signals. Besides, the number of cycles during repeatable programming, i.e., endurance, is another important metrics of PCMs. According to the above requirements, perfect PCMs for photonic devices with microheater fall into the intersectional zone of electronic phase-change memories and optical phase-change memories. Benefiting from the deep research on PCRAM and phase-change optical disks, PCMs with sharp electrical/optical contrast, while maintaining fast switching speed, are mainly located along the GeTe–Sb_2Te_3 pseudo-binary line with compositions $(GeTe)_m(Sb_2Te_3)$ [18]. One of the most well-known PCMs from this group is obviously the GST alloy, which has been widely implemented in various phase-change memories ranging from conventional rewritable optical disks to novel all-photonic memories. One common crystal structure of the GST alloy is the metastable rocksalt structure. It has Te atoms taking up sites on one face-centered-cubic (f.c.c.) sublattice and Ge and Sb randomly forming the other f.c.c. sublattice [19]. Such structure is also accompanied with numerous vacancies for the Ge/Sb sites. Note that the number of vacancies in the crystalline phase strongly differs from that in amorphous phase. The vacancies that give rise to p–p orbitals in Ge/Sb materials were vulnerable to local distortions. Such p–p orbitals can be misaligned due to heat-induced lattice vibrations during amorphization, which decreases the number of resonant bonds and consequently changes the optical properties of the amorphous state.

Despite its intensive study and widespread applications, GST alloy is still facing some insurmountable challenges. One severe drawback arises from its pronounced optical loss caused by high free carrier's absorption [20]. Besides, nonvolatile data retention period of GST is still shorter than the requirement of storing data for more than ten years. Accordingly, considerable research enthusiasm has turned to an improved version of GST, i.e., Ge–Sb–Se–Te (GSST) alloy. It is well known that chalcogenide alloy with Se element usually shows better optoelectronic properties and excellent glass-forming capability [21]. The inherent drawbacks of Se-based alloys, such as low sensitivity and aging effect, can be further mitigated by combining with Te elements. Given this fact, the isoelectronic substitution of Te by Se benefits the bandgap tuning and interband absorption [20]. More excitingly, the physical performances of Se-based alloy can be alleviated by incorporating additional dopants, such as Ge and Sb. The introduction of these dopants can effectively facilitate thermal stability and glass-forming region, while suppressing aging effect [21]. Owing to the above factors, such quaternary alloy when compared to its ternary opponents enabled fast switching speed, better thermal stability, and broadened far-infrared transmittance region, thus leading to a high figure-of-merit (FOM) [22] that refers to the performance evaluation index of phase-change memory. FOM serves as a comprehensive metric for assessing various aspects of device performance, including phase-change speed, energy efficiency, polymorphic behavior, optical characteristics, and cyclic stability. When employing FOM for the evaluation of device performance, a higher numerical value typically indicates superior performance. To assess optical characteristics, FOM is defined as the ratio of the change in refractive index (RI) (Δn) to the change in extinction coefficient (Δk) during a phase change (FOM $= \Delta n/\Delta k$). In this context, an elevated FOM signifies a more pronounced optical contrast in the device. Accordingly, the compositions from the GSST system have most recently been considered as more promising for photonic memory applications than from either Ge–Sb–Se or Ge–Sb–Te systems.

Although aforementioned photonic materials exhibit strong optical contrast, their absorption in crystalline phase is still large, which causes nonnegligible optical loss. Such loss requires frequent signal amplification when device cascading is demanded particularly for neural network applications.

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TABLE I
ELECTRICAL RESISTIVITY (ρ) AND COMPLEX RI AT THE WAVELENGTH OF 1550 NM OF COMMON PHOTONIC PCMS

PCMs	ρ(am)	ρ(cr)	Ri (am)	Ri (cr)	Cycles	Switch area
$Ge_2Sb_2Te_5$	30(Ω cm) [28] at 20°C	10^{-3}(Ω cm) [28] at 250°C	4.6+0.18i [9]	7.2+1.9i [9]	150 [33]	10^{-2}um^2 [33]
$Ge_2Sb_2Se_4Te_1$	500(Ω cm) [28] at 20°C	10^{-2}(Ω cm) [28] at 370°C	3.325+1.8×10^{-4}i [9]	5.083+0.35i [9]	40 [33]	10^5um^2 [33]
Sb_2S_3	10^9(Ω cm) [29] at 20°C	10^7(Ω cm) [30] at 250°C	2.712+0i [9]	3.308+0i [9]	1000 [33]	10^4um^2 [33]
Sb_2Se_3	10^6(Ω cm) [31] at 20°C	500(Ω cm) [32] at 200°C	3.285+0i [9]	4.050+0i [9]	5000 [33]	10^4um^2 [33]

PCMs with favorable properties were initially attributed to "resonance bonding" in the crystalline state [23]. However, such speculation was challenged by an emerging concept of "metavalent bonding" (MVB) [24]. MVB ascribed to the source of properties variations between crystalline and amorphous phases to the competition between electron delocalization as in metallic bonding and electron localization as in covalent or ionic bonding. An adversary theory proposed the existence of "partial bonding" in PCMs with two types of covalent bonds [25]. Further optimization can be thus accomplished by replacing Ge with Sb to lower the electronic polarizability and consequently RI, while substituting Te for Se or S to enhance the bandgap energy. Such strategies resulted in the emerging of two novel PCMs, i.e., Sb_2Se_3 and Sb_2S_3.

They were reported to allow for very low losses in the telecommunications *C*-band [26], and Sb_2S_3 exhibited stronger absorption in the visible spectrum [27]. Table I summarized the electrical and optical properties of common photonic PCMs.

III. STATE-OF-THE-ART PHASE-CHANGE INTEGRATED PHOTONIC DEVICES WITH MICROHEATERS

Note that microheater materials play an important role in determining physical performances of phase-change integrated photonic device with electrical excitation. Similar to resistive heater of PCRAM, the electrical resistivity of microheater needs to be fairly low in order to generate adequate Joule heating for phase transformation. The thermal conductivity of microheater is also required to be relatively low to confine heat at the interfacial region, while maintaining low optical loss. ITO is one of the common microheater materials due to its ability to satisfy the above requirements associated with the easy integration. One typical design was to form an ITO conductive bridge connecting two metal pads and to deposit GST on this conductive bridge, as illustrated in Fig. 3(a) [34]. Such design allowed for different crystallization levels by choosing appropriate electric pulses. This can lead to different transmissions and induce a variety of phase shifts. Similar architecture was also adopted using Au metal pads, as illustrated in Fig. 3(b) [35]. Injecting electrical pulse can electrically switch GST between full amorphous and crystalline phases, enabling optical switching functionality. In addition to GST alloy, electrically exciting phase transformation of Sb_2S_3 was achieved through a microring structure, as demonstrated in Fig. 3(c) [36]. An ITO microheater was integrated on top of Sb_2S_3, and the electrical excitation was applied between two Au/Cr electrodes to induce the phase transformation. It was clearly observed that the switching between "SET" and "RESET" can modulate the resonance tip from 1551.6 to 1552 nm. It is necessary to notice that the above designs were only devised at device/cell level, whereas more practical designs that can work at system level were still missing. To overcome this issue, crossbar electrodes that were widely adopted for PCRAM technology were embedded inside the SOI platform, and row and column electrodes were connected to two sides of directional couplers, where GST were deposited on. Due to this, any GST cells can be programmed by applying bias to appropriate electrodes, realizing random access functionality. Multilevel transmission levels were also obtained by continuously tailoring the crystallization extent, as shown in Fig. 3(d) [37].

In addition to ITO, graphene microheater has attained considerable attention owing to its proper electrical and in-plane thermal conductivities, ultralow heat capacity, good transparency, and compatibility with CMOS process [38]. Its superiority was theoretically demonstrated by capping GST with a single graphene layer, deposited on SOI platform, as illustrated in Fig. 4(a) [39]. Such design led to a high switching speed of ~80 MHz and a high energy efficiency of 19.2 aJ/nm^3 (6.6 aJ/nm^3) for crystallization (amorphization). Physical fabrication of phase-change photonics devices with graphene microheater was also performed by means of GSST material, reflected in Fig. 4(b) [40]. Using one or two amorphization pulses and two or three crystallization pulses enabled four different crystallization extents. Such four pronounced states can be distinguished by detecting the Raman signal difference of both the amorphous and crystalline signature peaks after each switching event. Similar architecture was also adopted by [41], which patterned a single-layer graphene bridge on SOI waveguide. The GST layer, located on the graphene bridge, can be electrically switched through the current applied between two Ti/Au electrodes, as shown in Fig. 4(c). This strategy was also applied to a microring-based phase shifter that replaced GST with Sb_2Se_3 [Fig. 4(d)]. Phase transformation of the Sb_2Se_3 via graphene microheater led to a maximum resonance shift of 0.021 nm, corresponding to a phase shift of 0.00822π. Most recently, an optical logic architecture, consisting of N-1 GSST-based compact two-waveguide based switches, optical-absorption switches, and N-input combiners, was proposed, as illustrated in Fig. 4(e) [42]. The fundamental switching unit in such optical logic architecture comprised a GSST-based

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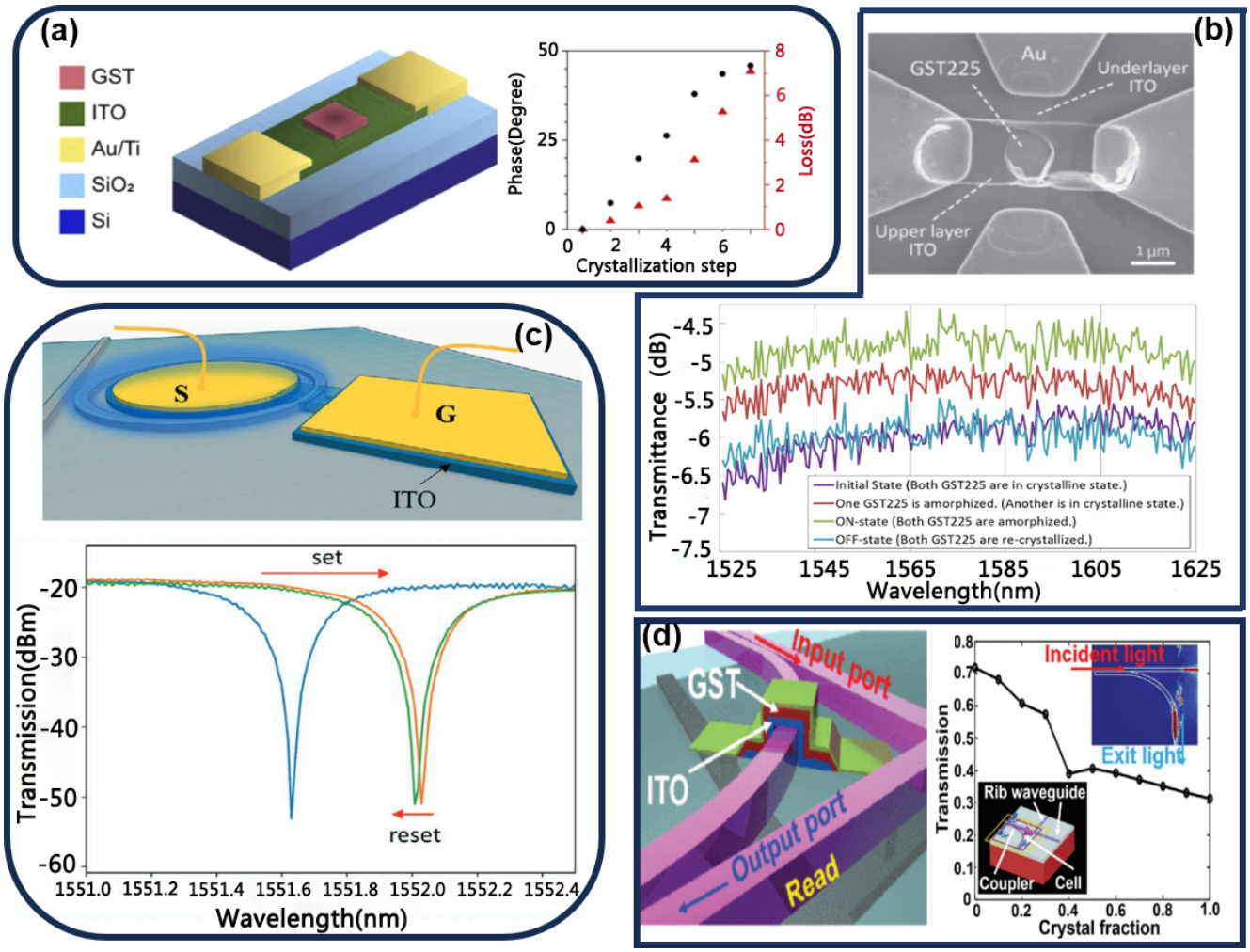


Fig. 3. (a) Schematic of the ITO microheater (left) and resulting phase shifts and optical loss with respect to different crystallization levels (right). (b) Scanning electron microscopy image of the proposed device (top) and transmittance spectra along with different GST phases (bottom). (c) Schematic illustration of the microring switching with ITO microheater (top) and measured transmission spectrum during "SET" and "RESET" processes (bottom). (d) Designed directional coupler with ITO microheater (left) and calculated device transmission along with different crystal fraction (right). (a)–(d) were reprinted with permissions from [34], [35], [36], and [37], respectively.

broadband electro-optic hybrid waveguide and a graphene microheater. Graphene microheater allowed GSST to be electrically switched between amorphous phase (i.e., logic 1) and crystalline phase (i.e., logic 0), thus governing the transmission in the output port, which is considered as output logic. This design provided for the realization of all the fundamental logic operations.

Single-doped Si and p-type/intrinsic/n-type (p-i-n) junctions are another two prospective microheater materials, which can be ascribed to their analogous foundry-compatible processes to active Si photonics devices. An optical switching, comprising a multimode interferometer (MMI) crossing with a nanoscale GST on top, was fabricated, resulting in Fig. 5(a) [15]. The crossing strip, made of heavily P^{++}-doped Si, was located at the center of the MMI, and a circular GST layer was deposited on the doped Si microheater. The electrical pulse applied to the doped Si layer induced resistive heating that caused phase transformation. Five different transmission levels were attained. Most recently, an in-memory photonic–electronic dot-product engine, composed of an array of nonvolatile GST memory cells with heavily P^{++}-doped microheaters, was proposed, as illustrated in Fig. 5(b) [43]. The input vectors were represented by the amplitudes of incident probe light, and output amplitudes resulted from the scalar multiplication of input vector and multiplicand that was mapped as the transmission of memory cells, which corresponded to their phase-transformation extents. Such strategy allowed for realization of convolutional processing for recognizing images. Phosphorus was also doped on a portion of the waveguide to serve as a n-doped microheater. Such microheater was incorporated with a bistable transparent PCM, Sb_2Se_3, to provide phase-shifting function [44], as depicted in Fig. 5(c). This phase shifter exhibited a remarkable phase modulation of up to 0.09 π/μ and maintained a low insertion loss of 0.3 dB/π. The single-doped microheater can be expanded arbitrarily to fit larger photonic device, advantageously differing from p-i-n case limited by the intrinsic size.

One typical phase-change photonic device that adopted p-i-n junction microheater was made use of a Si waveguide on a 100-nm-thick slab layer that was heavily doped by boron and phosphorus ion implantation. A thin GST film was subsequently deposited on the rib, and electrodes were made of Pd, as illustrated in Fig. 5(d) [45]. The electrically induced phase transformation gave rise to two distinct optical

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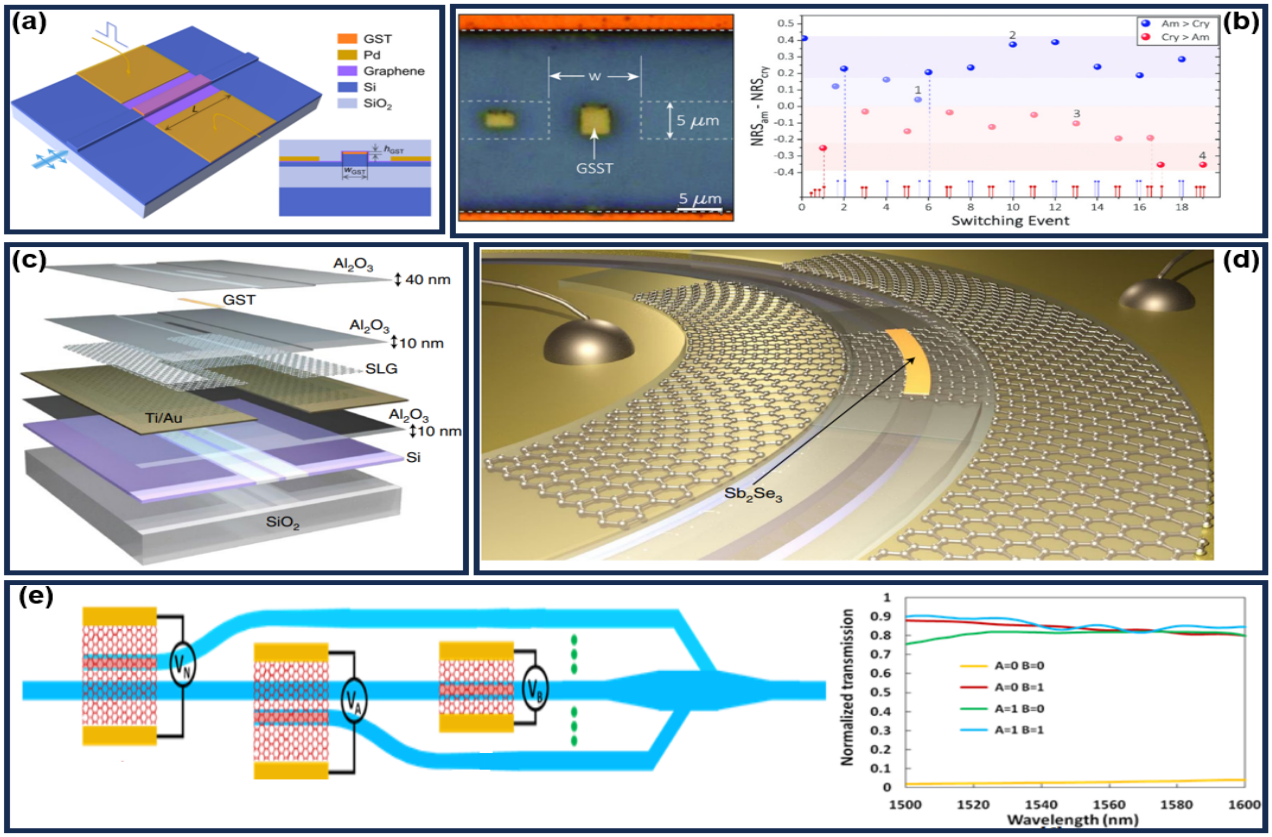


Fig. 4. (a) Schematic of the graphene microheater. (b) Optical microscopy image of the proposed device using GSST and graphene microheater (left) and measured peak height differences for normalized Raman signals (NRSs) for the amorphous and crystalline peaks at 159 and 120 cm^{-1}, respectively. (c) Layered architecture of the phase-change photonic device with graphene microheater. (d) Schematic of the graphene-Sb_2Se_3 phase shifter in a microring. (e) Schematic of the designed broadband nonvolatile N-bit logic circuit (left) and calculated normalized transmission spectra with respect to different input logics for "OR" function. (a), (b), and (e) were reprinted with permissions from [39], [40], and [42], while (c) and (d) were reprinted with permissions from [41].

output spectra after two reversible "SET" and "RESET" processes. The p-i-n microheater can also be integrated with a directional coupler coated by a GST strip [46], as shown in Fig. 5(e). Such design showcased remarkable switching cycles (>2800) and outstanding scalability. Besides, the p-i-n microheater found its appropriateness for three integrated photonic devices employing Sb_2S_3, i.e., microring resonators, Mach–Zehnder interferometer (MZI), and asymmetrical directional couplers [47], as illustrated in Fig. 6(a). In addition to GST and Sb_2S_3, a 2 × 2 MZI switch integrated with low-loss Sb_2Se_3 was proposed [48], resulting in Fig. 6(b). Its phase-change process was employed by a forward-biased p-i-n diode, revealing 6-bit multilevel switching states and more than 10 000 reversible cycles. Theoretical modeling of p-i-n microheater-based photonic device was also performed, as illustrated in Fig. 6(c) [49], and compared to single-doped Si and double microheaters. p-i-n heater-based device was reported to exhibit greater heating efficiency and higher operational speed.

Besides the above microheaters, recent progress was witnessed on metal microheater and mixed-model switching technologies. One innovative approach was recently reported by imposing alumina (Al_2O_3) between PCMs and tungsten (W)/titanium (Ti) metal microheater [50], as illustrated in Fig. 6(d). One benefit of choosing nonplasma-treated Ti for the heating element is its low optical loss, resulting in exceptional durability with up to 500 000 switching cycles. Nevertheless, metal heaters can be susceptible to physical damage, resulting from oxidation and repeated heating and cooling cycles. Differing from aforementioned designs, recently reported mixed-mode memory [51], resulting in Fig. 6(e), filled PCMs within the gap between electrodes and thus allowed for vast field enhancement for programming. Resulting resistance/transmission changes can be sensed electrically/optically. By decoupling the conduction of GST from the Joule heating mechanism, such design was operated with an exceptionally low switching voltage and enabled light transmission via electronic means on a nanoscale platform.

IV. Challenges and Prospects

Today, photonic–electronic hybrid hardware is receiving vast attention for neuromorphic and in-memory computing applications. Prior to PCMs, various materials, such as graphene, polymers, and electro-optic materials, have been considered and studied deeply for photonic–electronic applications. These candidates are, however, volatile that require electrical excitations to maintain their physical states. Given this fact, PCMs-based hybrid hardware obviously exhibits its performance superiorities particularly in terms of its nonvolatile property. PCMs can be either optically or electrically switched between amorphous and crystalline phases via free-space optical heating, ON-chip optical heating,

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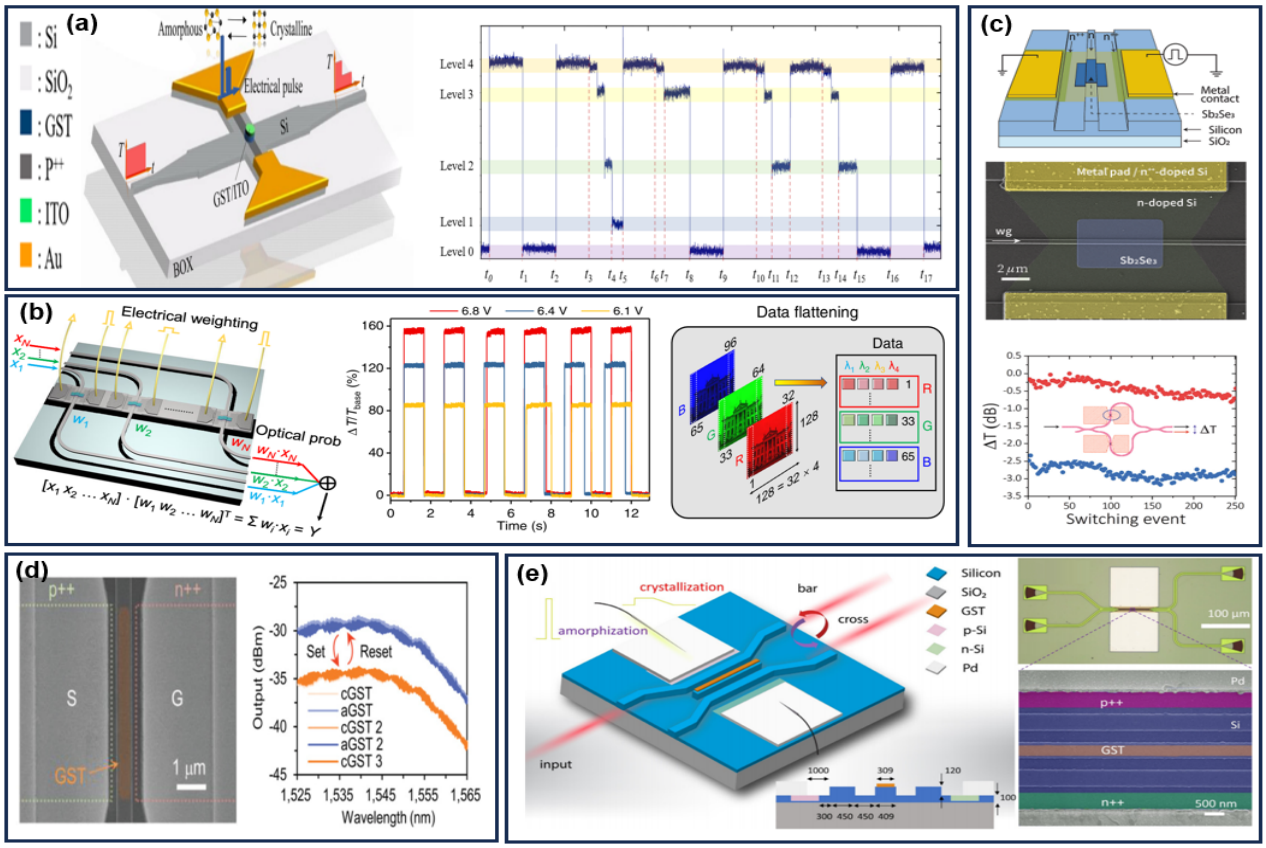


Fig. 5. (a) Schematic of the single-doped Si microheater (left) and measured transmission levels with respect to each partially crystallized extent (right). (b) Three-dimensional view and operational mechanism of the dot-product engine with p-i-n microheater (left) and its normalized transmission with respect to different phase-transformation extent, caused by different voltages (middle), and flattened pixel data of an RGB image encoded in amplitudes of four different wavelength channels by four variable optical attenuators (right). (c) Schematic of a microheater with doped silicon profile (top) and SEM image of the device (middle) and reversible switching over more than 125 periods between two intermediate states of the MZI upper-arm phase shifter. (d) SEM image of the GST with p-i-n microheater (left) and optical output spectra after two reversible "SET" and "RESET" processes (right). (e) Schematic of the broadband nonvolatile electro-optic 2 × 2 programmable device (left) and optical micrograph of the device (top right) along with SEM image (bottom right). (a)–(e) were reprinted with permissions from [15], [43], [44], [45], and [46], respectively.

and electrical threshold switching. Free-space optical heating undoubtedly deteriorates the device integration, whereas ON-chip optical heating has difficulty in several aspects, such as switching large volume and routing/generating optical signal. Conventional electrical threshold switching usually causes nonuniform heating and formation of conductive filaments, thus impairing the photonic applications. PCMs-based photonic–electronic hardware with external microheater can locally choose and arbitrarily extend the switching region through the interfacial heating, thus capable of circumventing the above limits. For the above reasons, designing a PCM-based photonic–electronic circuit with electrical programming and optical probe is of high importance for future neuromorphic and in-memory photonic computing.

It should be noticed that large optical contrast between amorphous and crystalline phases is compulsory to PCMs for photonic applications. Under these circumstances, the GST alloy that was widely adopted by rewritable disk and PCRAM is still a suitable option for phase-change photonic device with integrated microheater. In spite of its storage and computing merits, the main drawback of GST arises from its relatively large absorption in its amorphous phase [20]. This adversely restricts its size and scalability and caused significant optical loss. One prospective approach is to leverage GSST to replace GST. The GSST exhibited higher FOM and optical loss two orders of magnitude lower than conventional GST alloy [20]. This can be ascribed to the remarkably reduced free carrier concentration and mobility that gives rise to blue-shifted interband transitions and minimal free-carrier absorption [52]. Despite its attractive traits, GSST still presents relatively large absorption in its crystalline phase. This evidently gives rise to unnegligible optical loss when numerous devices are cascaded to construct neural networks. The limits of GSST can be further mitigated using a class of Sb-based binary compound, such as Sb_2S_3 and Sb_2Se_3. These type of alloys exhibits a sharp contrast of RI and very low losses in the telecommunications C-band. Resulting FOM from Sb_2Se_3 was also reported to be two orders of magnitude higher than GST case [26], implying their bright prospect for ON-chip storage and computing. Although optical loss was strongly suppressed by these novel photonic PCMs, their endurance still remained challenged particularly when compared with conventional PCMs for PCRAMs. Several materials and design criterions, including single phase system, minimal volume change during phase transition, temperature uniformity, and encapsulation design, were reported in order to greatly enhance the materials endurance [33]. Besides, some instances of PCMs-based photonic devices with microheater implemented

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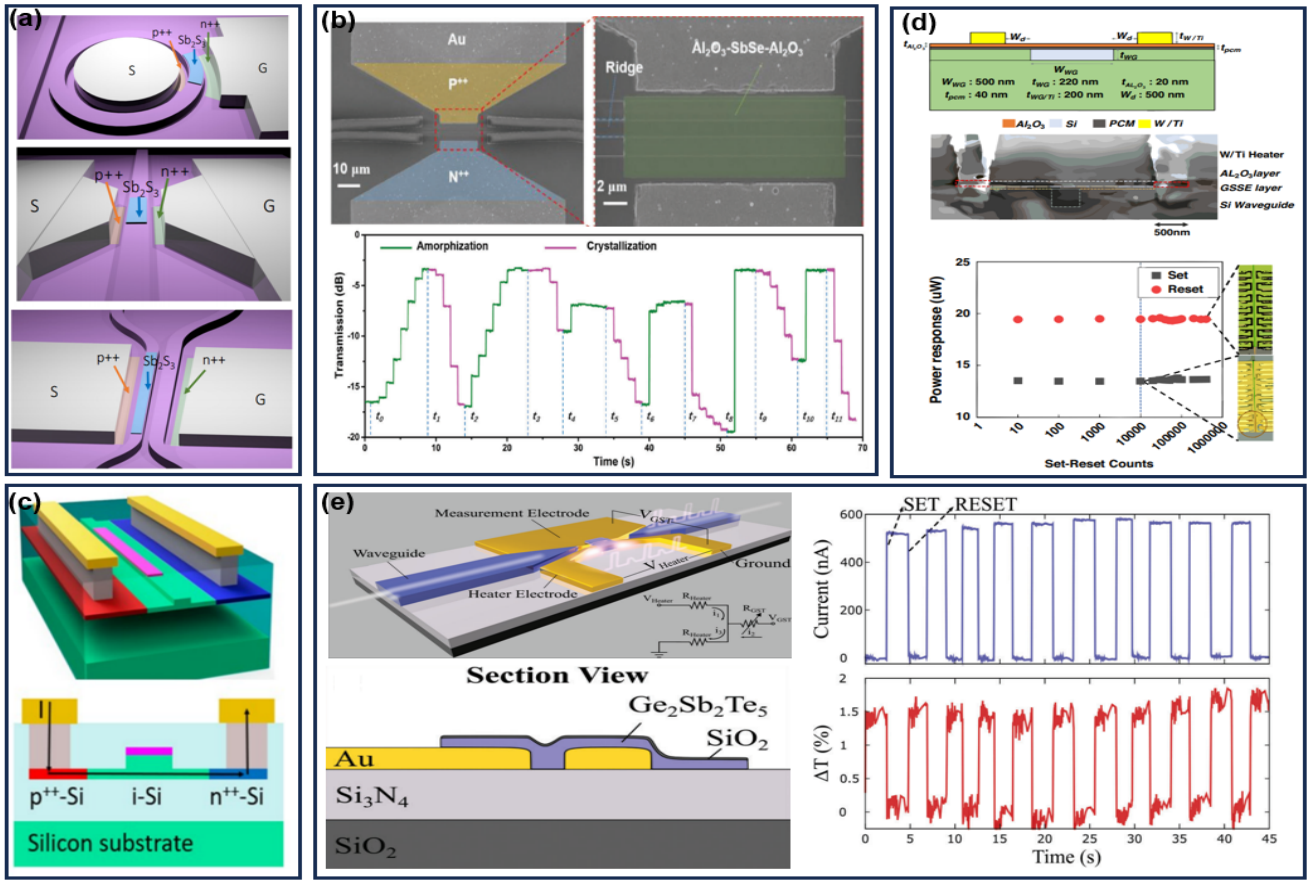


Fig. 6. (a) Schematics of the ring resonator device (top), Mach–Zehnder modulator device (middle), and Si–Sb_2S_3 hybrid waveguide asymmetric directional coupler device (bottom). (b) SEM image of the MZI switch (top) and the device's dynamic multilevel tuning process (bottom). (c) Three-dimensional diagram of the GST-based photonic device with p-i-n microheater (top) and its cross section (bottom). (d) Cross-sectional schematic of the device (top), SEM image depicting the device cross section (middle), and optical response variation observed over 500 000 switching cycles (bottom). (e) Schematic of the device includes Si_3N_4 optical waveguide, plasma MIM waveguide, and phase-change element (top left), cross-sectional view of the phase-change unit (bottom left), and continuous readout of current and light transmission on the PCM (right). (a)–(e) were reprinted with permissions from [47], [48], [49], [50], and [51], respectively.

very large switching voltage (>8 V) for amorphization [53], thus making energy consumption problematic. This can be alleviated by either reducing the PCMs volume or leveraging PCMs with lower melting points. Moreover, choosing appropriate microheater and encapsulation materials can benefit heat localization effect and thus reduce the switching voltage.

As Joule heating is transferred to PCMs through the microheater, its constitutions need to be selected carefully in order to optimize the heating efficiency. To date, the microheater components can be mainly classified into ITO, graphene, doped Si, and Si p-i-n junction. The performance comparison of photonic devices with different microheaters is given in Table II. One advantage of the ITO microheater can be ascribed to its low thermal conductivity that can effectively confine heat inside a localized region [34]. Moreover, depending on its size, the electrical conductivity of the ITO layer varies from 10^{-2} to 10^6 $\Omega^{-1}m^{-1}$, which can be thus modulated flexibly to provide adequate Joule heating. Nevertheless, an unnegligible insertion loss (<1 dB per component) for large-scale integrated components is still witnessed for ITO microheater that also shows relatively slow switching speed. This can be mitigated by precisely controlling the carrier concentration of ITO via ion-beam-assisted e-beam evaporation, which can modulate RI of ITO and thus reduce the loss [54]. Most recently, another transparent microheater, i.e., indium oxide (In_2O_3) enabled seamless integration with diverse photonic platforms. With an electrical conductivity of 10^6 $\Omega^{-1}m^{-1}$, In_2O_3 generated adequate Joule heating required for phase transition. In the telecom band, In_2O_3 thin films exhibited an effective RI of approximately $2 + 0.015i$, resulting in minimal disruption to optical modes [55]. Graphene microheater was reported to exhibit high plane electrical conductivity, ultralow heat capacity, and large optical transparency [40]. This reduced the optical loss to 0.07 dB/μm at 1550 nm for one layer graphene in association with much lower energy consumption and higher switching speed. Despite its promising traits, the presence of cracks and leaky grain boundaries inside the transferred graphene layer may severely exacerbate its mechanical fractures. Utilizing a polymer-assisted approach, it is feasible to achieve the transfer of graphene without the development of cracks, offering a promising solution for the large-scale transfer of graphene films [56]. For doped Si [15] and Si p-i-n junctions [45], both microheaters are seamlessly compatible with current Si photonic fabrication process. The fact that waveguide can be patterned in the intrinsic region of the junction endowed Si p-i-n microheater with very low optical loss down to 0.02 dB/μm. However, its heating efficiency strongly pertains to the doping level and

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TABLE II
PERFORMANCES COMPARISON AMONG DIFFERENT MICROHEATERS

Heater	PCMs	Switching mechanism	Switch time (a/c)	Energy consumption(a/c)	Switching energy	Microheater geometry	Modulation type	CMOS-backend compatibility	Endurance
ITO (Sim) [37]	GST	Electrical	100 ns/120 ns	1.2 nJ/0.96 nJ	105 aJ/nm^3	Rectangular	Amplitude modulation	Yes	–
ITO (Exp) [34]	GST	Electrical	50 ns/200 ns	10 nJ/4.4 mJ	111 aJ/nm^3	Bridge	Phase modulation	Yes	1
Graphene (Sim) [39]	GST	Electrical	0.22 ns/8 ns	0.066 nJ/1.92 nJ	10 aJ/nm^3	Rectangular	Amplitude modulation	Yes	–
Graphene (Exp) [40]	GSST	Electrical	13 µs/20 ms	117 nJ/120 µJ	485 aJ/nm^3	Bridge	Amplitude modulation	Yes	9
Graphene (Exp) [41]	Sb_2Se_3	Electrical	408 ns/220 µs	5.5 nJ/0.99 µJ	127.6 aJ/nm^3	Bridge	Phase modulation	Yes	1000
Doped Si (Exp) [15]	GST	Electrical	20 ns/200 ns	10 nJ/18 nJ	755 aJ/nm^3	Rectangular	Amplitude modulation	Yes	50
Doped Si (Exp) [44]	Sb_2Se_3	Electrical	400 ns/1 ms	66 nJ/9.6 µJ	92 aJ/nm^3	Bridge	Phase modulation	Yes	125
PIN (Exp) [43]	GST	Electrical	50 ns/200 ns	8.8 nJ/6.9 nJ	12.8 aJ/nm^3	Bridge	Amplitude modulation	Yes	100
PIN (Exp) [45]	GST	Electrical	100 ns/50 µs	8 nJ/30 nJ	728 aJ/nm^3	Bridge	Amplitude modulation	Yes	500
In_2O_3(Exp) [55]	GST	Electrical	2.8 us/100 µs	80 nJ/120 nJ	5300 aJ/nm^3	Rectangular	Amplitude modulation	Yes	1
W/Ti (Exp) [50]	GSSe	Electrical	2 us/20 µs	80 nJ/400 nJ	164 aJ/nm^3	Rectangular	Amplitude modulation	Yes	500000
Laser(Exp) [58]	Sb_2Se_3	Optical	400 ns/50 ms	14 nJ/950 µJ	2.09 x 10^5 aJ/nm^3	–	Phase modulation	No	500
Laser(Exp) [59]	GST	Optical	25 ns/15 us	180 pJ/ 17 nJ	1900 aJ/nm^3	–	Amplitude modulation	No	1000

device geometry, thus increasing the difficulty in performance optimization. In contrast, the device design of doped Si microheater is relatively straightforward and simple, allowing for relatively large size but at the expense of high energy consumption. It was reported that in the P-doped region of p-i-n heaters, devices with smaller doped areas consistently outperformed those with larger doped regions [57]. Moreover, for applications with ON-chip voltages below 5 V, devices with higher doping concentrations can elevate temperatures at lower voltage levels.

Device robustness is one of the most important performance metrics, mainly affected by properties and geometrical variations among different devices. One promising approach to mitigate robustness is to adopt plasmonic resonance structures by filling PCMs within the gap between two plasmonic metal disks. Resulting plasmonic resonance effect remarkably enhanced electric field and thus induces phase transformation. As such method exhibits less dependence on property/geometrical variations [10], leveraging plasmonic enhancement pathways not only facilitates the realization of faster and more energy-efficient phase-change integrated photonic devices but also enhances manufacturing tolerance and suppresses device-to-device variations [10]. Moreover, as nanodevice fabrication technologies continue to advance, it is anticipated that device performance differentials stemming from manufacturing variances will progressively diminish and ultimately vanish.

V. CONCLUSION

Phase-change photonic technologies with integrated microheater was reviewed here in terms of their physical mechanisms, photonic materials, and microheater design. The-state-of-the-art phase-change photonic devices was also discussed according to different microheater constitutions, and their respective merits and weakness were eventually analyzed associated with their future application potential.

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